

# Machine Learning

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October 19, 2021

# Introduction

- ▶ Age of automation and intelligent systems
- ▶ Enormous potential accross industries, enterprises
- ▶ Intelligent systems and data-driven organizations are becoming a reality
- ▶ Advancements in tools and techniques is only helping it expand further

**Data is the new oil and Machine Learning is a powerful concept and framework for making the best out of it.**

# Introduction

## Aim of this course (1)

- ▶ The core idea is to give you enough background on why we need Machine Learning,
- ▶ The fundamental building blocks of Machine Learning,
- ▶ What Machine Learning offers us presently.
- ▶ This will enable you to learn about how best you can leverage Machine Learning to get the maximum from your data.

# Introduction

## Aim of this course (2)

- ▶ Understand formal definitions,
- ▶ Concepts, foundations with regard to learning algorithms,
- ▶ Data management,
- ▶ Model building, evaluation, and deployment.

# Introduction

## Aim of this course (3)

### Practical aspect of the course

- ▶ Specific use cases,
- ▶ Specific problem,
- ▶ Real-world case.

# ML Basics

## The need for Machine Learning

Why make machine learn?

- ▶ Lack of sufficient human expertise in a domain,
- ▶ Scenarios and behavior can keep changing over time,
- ▶ Humans have sufficient expertise in the domain but it is extremely difficult to formally explain or translate this expertise into computational tasks,
- ▶ Addressing domain specific problems at scale with huge volumes of data.

# ML Basics

## Traditional Programming Paradigm

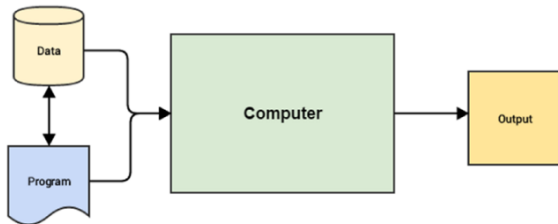


Figure 1: Traditional Programming Paradigm

# ML Basics

## Why Machine Learning?

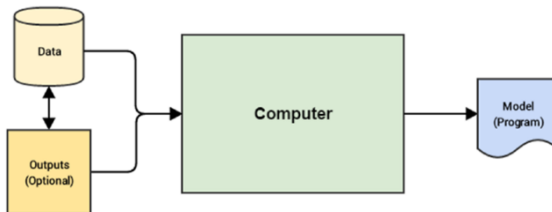


Figure 2: ML Programming Paradigm



# ML Basics

## Solve Machine Learning problem

- ▶ Leverage device data and logs,
- ▶ Decide key data attributes that could be useful for building a model,
- ▶ Observe and capture device attributes and their behavior over various time periods,
- ▶ Feed these input and output pairs to any specific Machine Learning algorithm,
- ▶ Deploy this model such that for newer values of device attributes it can predict if a specific device is behaving normally or it might cause a potential outage.

# ML Basics

## General definition

The need for Machine Learning

- ▶ Making Data-Driven Decisions
- ▶ Efficiency and Scale

**Machine Learning** is the field of study that gives computers the ability to learn without being explicitly programmed. *Arthur Samuel, 1959*

# ML Basics

## Formal definition

*A computer program is said to learn from experience  $E$  with respect to some class of tasks  $T$  and performance measure  $P$ , if its performance at tasks in  $T$ , as measured by  $P$ , improves with experience  $E$*

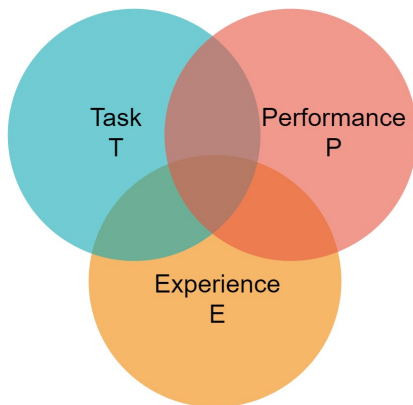


Figure 3: ML Ven Diagramme

# Task, Experience & Performance

*A task,  $T$ , can usually be defined as a Machine Learning task based on the process or workflow that the system should follow to operate on data points or samples.*

- ▶ Classification or categorization
- ▶ Regression
- ▶ Anomaly Detection
- ▶ Structured annotation
- ▶ Translation
- ▶ Clustering or grouping
- ▶ Transcriptions

# Task, Experience & Performance

The process of consuming a dataset that consists of data samples or data points such that a learning algorithm or model learns inherent patterns is defined as **the experience,  $E$**  which is gained by the learning algorithm.

**The performance,  $P$** , is usually a quantitative measure or metric that's used to see how well the algorithm or model is performing the task,  $T$ , with experience,  $E$ .

# Machine Learning: a true multi-disciplinary field

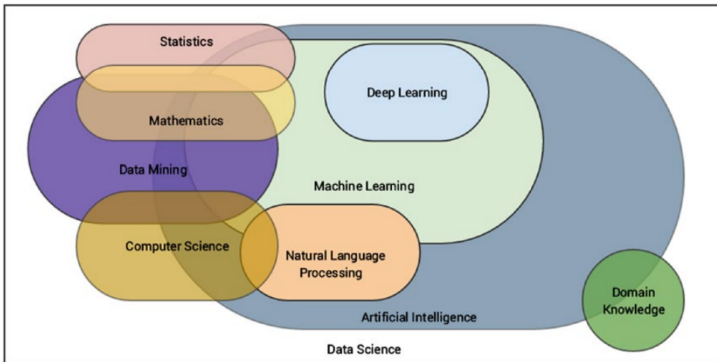


Figure 4: Discipline Ven Diagramme

# (Artificial Intelligence)

The art, science and engineering of making intelligent agents, machines and programs.

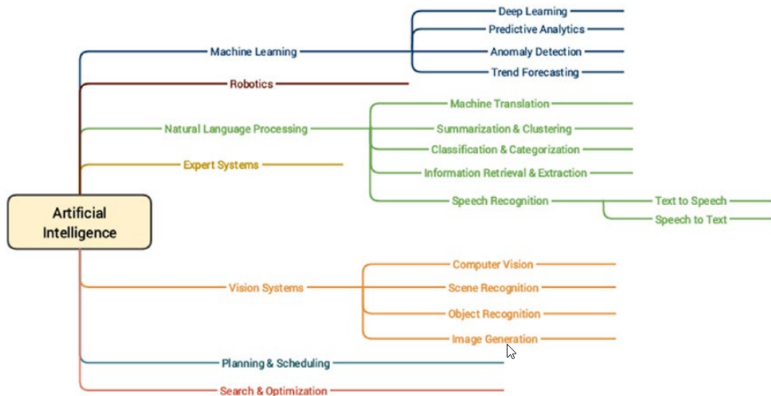


Figure 5: Diverse major facets under the AI umbrella

# ML Basics

## Deep Learning

Deep Learning based approach tries to build machine intelligence by representing data as a layered hierarchy of concepts, where each layer of concepts is built from other simpler layers.

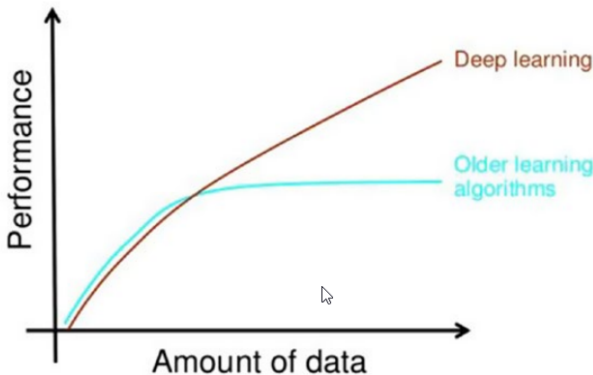


Figure 6: Performance comparison of Deep Learning and traditional Machine Learning by Andrew Ng



# ML Basics

## Important Concepts of DL

- ▶ Artificial Neural Networks
- ▶ Backpropagation
- ▶ Multilayer Perceptrons
- ▶ Convolutional Neural Networks
- ▶ Recurrent Neural Networks
- ▶ Long Short-Term Memory Networks
- ▶ Autoencoders

# ML Basics

## Artificial Neural Networks illustration

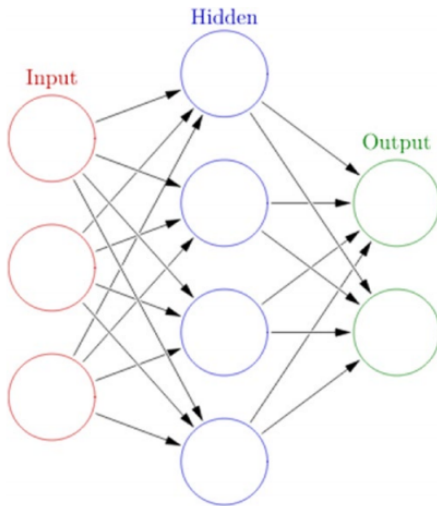


Figure 7: A typical artificial neural network

# Machine Learning Methods

1. **Methods based on the amount of human supervision in the learning process**
  - ▶ Supervised learning
  - ▶ Unsupervised learning
  - ▶ Semi-supervised learning
  - ▶ Reinforcement learning
2. **Methods based on the ability to learn from incremental data samples**
  - ▶ Batch learning
  - ▶ Online learning
3. **Methods based on their approach to generalization from data samples**
  - ▶ Instance based learning
  - ▶ Model based learning

# Supervised Learning

Supervised learning methods or algorithms include learning algorithms that take in data samples (known as training data) and associated outputs (known as labels or responses) with each data sample during the model training process. The main objective is to

learn a mapping or association between input data samples  $\mathbf{x}$  and their corresponding outputs  $\mathbf{y}$  based on multiple training data instances. Main supervised learning methods :

- ▶ Classification
- ▶ Regression

# Supervised Learning : Classification

predict output labels or **classes** or responses that are categorical in nature for input data based on what the model has learned in the training phase.

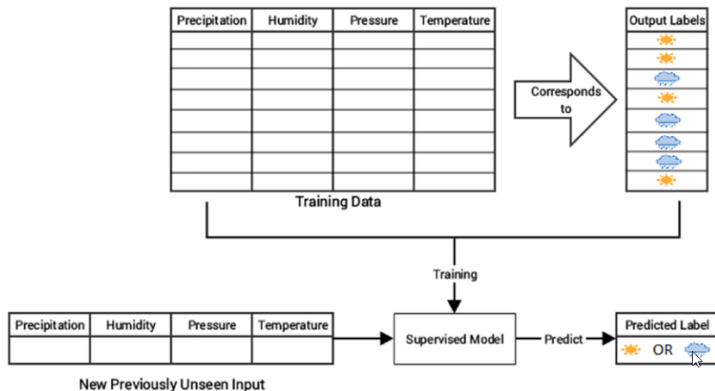


Figure 8: Illustration of Classification

# Supervised Learning : Regression

Machine Learning tasks where the main objective is value estimation can be termed as *regression tasks*.

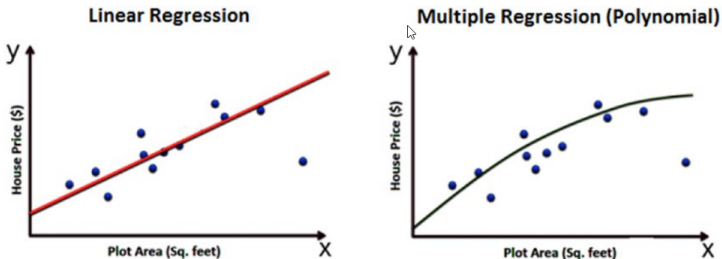


Figure 9: Supervised learning: regression models for house price prediction

# Unsupervised Learning

the model or algorithm tries to learn inherent latent structures, patterns and relationships from given data without any help or supervision. Unsupervised learning methods can be categorized under the following broad areas of ML tasks

- ▶ Clustering
- ▶ Dimensionality reduction
- ▶ Anomaly detection
- ▶ Association rule-mining

# Unsupervised Learning : Clustering

Clustering methods are Machine Learning methods that try to find patterns of similarity and relationships among data samples in our dataset and then cluster these samples into various groups, such that each group or cluster of data samples has some similarity, based on the inherent attributes or features.

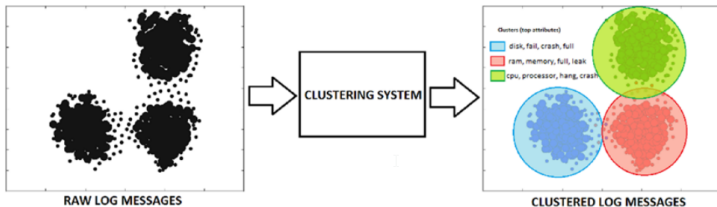


Figure 10: Unsupervised learning: clustering log messages



# Unsupervised Learning : Dimensionality Reduction

These methods reduce the number of feature variables by extracting or selecting a set of principal or representative features.

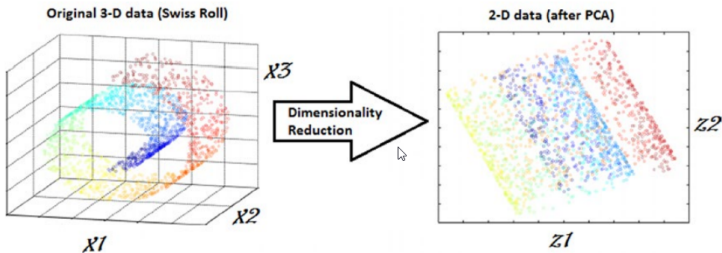


Figure 11: Unsupervised learning: dimensionality reduction

# Unsupervised Learning : Anomaly Detection

We are interested in finding out occurrences of rare events or observations that typically do not occur normally based on historical data samples.

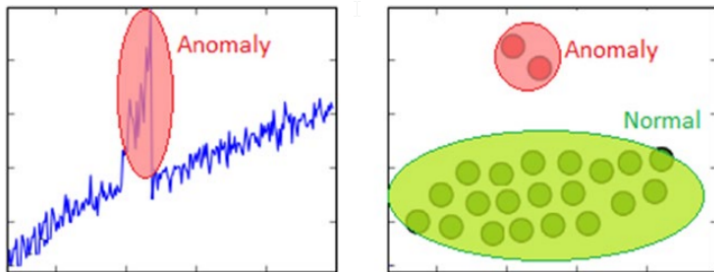


Figure 12: Unsupervised learning: anomaly detection

# Unsupervised learning : Association Rule-Mining

Typically association rule-mining is a data mining method use to examine and analyze large transactional datasets to find patterns and rules of interest.

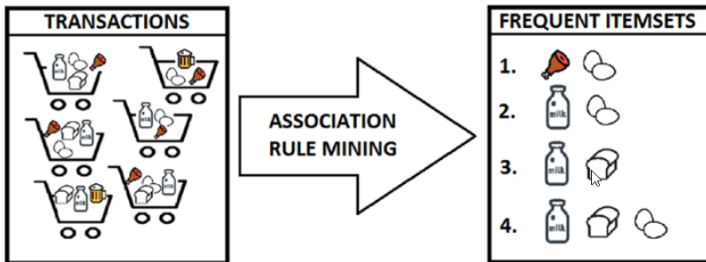


Figure 13: Unsupervised learning: association rule-mining

# Semi-Supervised Learning : Reinforcement Learning

We have an agent that we want to train over a period of time to interact with a specific environment and improve its performance over a period of time with regard to the type of actions it performs on the environment.

1. Prepare agent with set of initial policies and strategy
2. Observe environment and current state
3. Select optimal policy and perform action
4. Get corresponding reward (or penalty)
5. Update policies if needed
6. Repeat Steps 2 - 5 iteratively until agent learns the most optimal policies

# Semi-Supervised Learning : Reinforcement Learning

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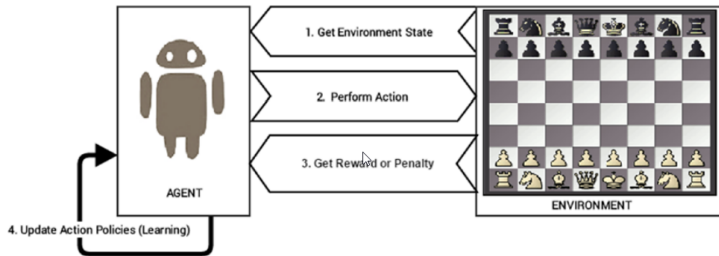


Figure 14: Reinforcement learning: training a robot to play chess

# Machine Learning overview

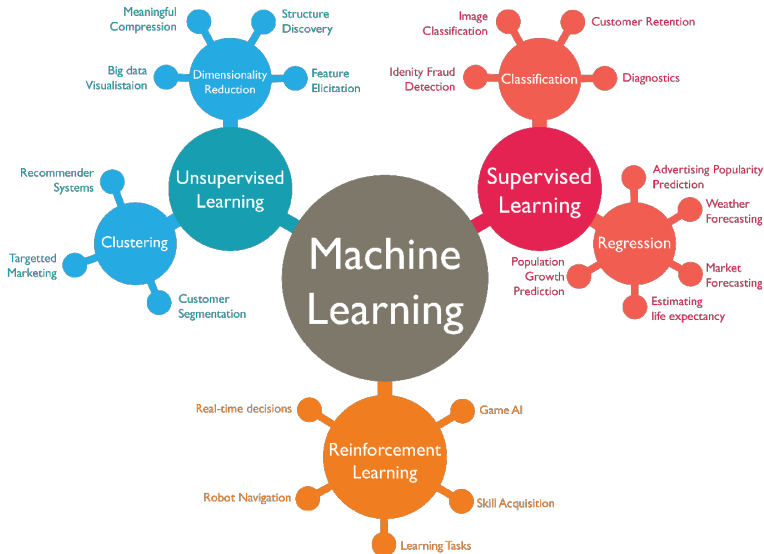


Figure 15: Machine learning overview

# Machine Learning pipeline

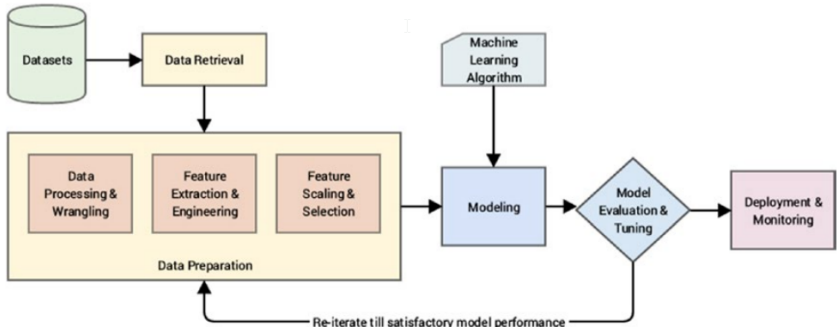


Figure 16: Machine learning pipeline

# Machine Learning Challenges (1)

- ▶ Data quality issues lead to problems,
- ▶ Data acquisition, extraction, and retrieval is an extremely tedious and time consuming process,
- ▶ Lack of good quality and sufficient training data in many scenarios,
- ▶ Formulating business problems clearly with well-defined goals and objectives,



## Machine Learning Challenges (2)

- ▶ Feature extraction and engineering, especially hand-crafting features,
- ▶ Overfitting or underfitting models,
- ▶ The curse of dimensionality: too many features can be a real hindrance,
- ▶ Complex models can be difficult to deploy in the real world.

# Real-World Applications of Machine Learning (1)

- ▶ Product recommendations in online shopping platforms,
- ▶ Sentiment and emotion analysis,
- ▶ Anomaly detection,
- ▶ Fraud detection and prevention,
- ▶ Content recommendation (news, music, movies, and so on)

## Real-World Applications of Machine Learning (2)

- ▶ Weather forecasting
- ▶ Stock market forecasting
- ▶ Market basket analysis
- ▶ Customer segmentation
- ▶ Object and scene recognition in images and video

# Real-World Applications of Machine Learning (3)

- ▶ Speech recognition
- ▶ Churn analytics
- ▶ Click through predictions
- ▶ Failure/defect detection and prevention
- ▶ E-mail spam filtering

# Python ML Ecosystem

## Definition

**Python** is an interpreted language, which means that the source code of a Python program is converted into bytecode, which is then executed by the Python virtual machine.

- ▶ Python code is fast to develop.
- ▶ Python code is not as fast in execution.

# Python ML Ecosystem

## Strengths

- ▶ Easy to learn
- ▶ Supports multiple programming paradigms
- ▶ Extensible
- ▶ Active open source community

## Setting Up a Python Environment

- ▶ Install Python and the necessary libraries individually
- ▶ Use a pre-packaged Python distribution that comes with necessary libraries, i.e. Anaconda

# Python ML Ecosystem

## Setting Up Anaconda Env

- ▶ Installing packages
- ▶ Jupyter notebook
- ▶ numpy
- ▶ Pandas
- ▶ Scikit-learn
- ▶ Theano
- ▶ Tensorflow
- ▶ Keras
- ▶ NLTK (The Natural Language Tool Kit)

# Data Engineering

## Intro

- ▶ **Data collection:** To understand different data retrieval mechanisms for different data types
- ▶ **Data description:** To understand various attributes and properties of the data collected
- ▶ **Data wrangling:** To prepare data for consumption in the modeling steps
- ▶ **Data visualization:** To visualize different attributes for sharing results, better understanding, and so on



# Regression

## Intro

### Problem

You want to train a model that represents a linear relationship between the feature and target vector.

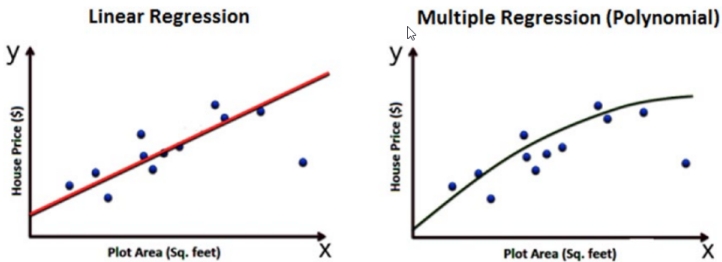


Figure 17: Supervised learning: regression models for house price prediction

# Regression

## Covers

- ▶ Visualization of data
- ▶ Simple linear regression implementation using normal equations (using Numpy)
- ▶ Gradient descent optimization (using numpy)
- ▶ R-squared implementation using numpy
- ▶ Residual plot analysis
- ▶ Comparison of model using normal equation with scikit implementation

# Regression

Given  $X_1, X_2, \dots, X_d$  (*predictor, explanatory, or independent variables*), and given  $Y$  (*response or dependent variable*), *regression* aims to predict  $Y$  based on  $X$ .

That is, the goal is to learn a *regression function*  $f$ , such that

$$Y = f(X_1, X_2, \dots, X_d) + \varepsilon = f(X) + \varepsilon$$

where  $X = (X_1, X_2, \dots, X_d)^T$  is the multivariate random variable comprising the predictor attributes, and  $\varepsilon$  is a random *error term* that is assumed to be independent of .

$Y$  is comprised of two components, one dependent on  $X$ , and the other, coming from the error term, independent of the predictor attributes.

The error term encapsulates inherent uncertainty in  $Y$ , as well as, possibly the effect of unobserved, hidden or *latent* variables.

# Linear Regression

In *linear regression* the function  $f$  is assumed to be linear in  $X$ , that is

$$f(X) = \beta + w_1X_1 + w_2X_2 + \cdots + w_dX_d = b + \sum_{i=1}^d w_iX_i = b + bw^T X \quad (1)$$

$b$  is the true (unknown) *bias* term,  $w_i$  is the true (unknown) *regression coefficient* or *weight* for attribute  $X_i$ , and  $w = (w_1, w_2, \cdots, w_d)^T$  is the true  $d$ -dimensional weight vector.

$f$  specifies a hyperplane in  $d+1$ , where  $w$  is the the weight vector that is normal or orthogonal to the hyperplane, and  $b$  is the *intercept* or offset term.

$f$  is completely specified by the  $d + 1$  parameters comprising  $b$  and  $w_i$ , for  $i = 1, \cdots, d$ .

## Linear regression

A common approach to predicting the bias and regression coefficients is to use the method of *least squares*.

Given the training data  $D$  with points  $x_i$  and response values  $y_i$  (for  $i = 1, \dots, n$ ), we seek values  $b$  and  $w$ , so as to minimize the sum of squared residual errors (SSE)

$$SSE = \sum_{i=1}^n \epsilon_i^2 = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \sum_{i=1}^n (y_i - b - w^T x_i)^2 \quad (2)$$

In bivariate regression,  $x$  comprises a single predictor attribute,  $X = (x_1, x_2, \dots, x_n)^T$ , along with  $Y = (y_1, y_2, \dots, y_n)^T$ :

$$y_i = f(x_i) = b + w \cdot x_i$$

# Regression

## Implementation : Basic steps

- ▶ Import the packages and classes you need.
- ▶ Provide data to work with and eventually do appropriate transformations.
- ▶ Create a regression model and fit it with existing data.
- ▶ Check the results of model fitting to know whether the model is satisfactory.
- ▶ Apply the model for predictions.

# Classification

## Intro

**The classification task is to predict the label or class for a given unlabeled point.**

Formally, a classifier is a model or function  $M$  that predicts the class label  $\hat{y}$  for a given input example  $x$ , that is,  $\hat{y} = M(x)$ , where  $\hat{y} \in c_1, c_2, \dots, c_k$  and each  $c_i$  is a **class, label** (a categorical attribute value).

- ▶ Classification is a supervised learning approach, since learning the model requires a set of points with their correct class labels, which is called a training set.
- ▶ After learning the model  $M$ , we can automatically predict the class for any new point.

# Classification

## Outline

- ▶ Probabilistic Classifier
- ▶ Decision Tree Classifier
- ▶ Linear Discriminant Analysis
- ▶ Support Vector Machine (SVM)
- ▶ Classification Assessment



# Classification

## Probabilistic Classifier

**Classification refers to the task of predicting a class label for a given unlabeled point.**

- ▶ **The (full) Bayes classifier** : which uses the Bayes theorem to predict the class as the one that maximizes the posterior probability.
- ▶ **The naive Bayes classifier** : which assumes that attributes are independent, but it is still surprisingly powerful for many applications.
- ▶ **The nearest neighbors classifier** : which uses a non-parametric approach to estimate the density.

# Classification

## Probabilistic Classifier : Bayes Classifier

### Problem

- ▶  $D$  : The training Dataset consist of  $n$  points  $x_i$  in a  $d$ -dimensional space.
- ▶  $y_i$  denote the class for each point, with  $y_i \in c_1, c_2, \dots, c_k$  .

The Bayes classifier directly uses the Bayes theorem to predict the class for a new test instance,  $x$ . It estimates the posterior probability  $P(c_i|x)$  for each class  $c_i$  , and chooses the class that has the largest probability. The predicted class for  $x$  is given as

predicted class for  $x$

$$\hat{y} = \operatorname{argmax}_{c_i} P(c_i|x)$$

# Classification

Probabilistic Classifier : Naive Bayes Classifier

## NBC

The naive Bayes approach makes the simple assumption that all the attributes are independent. This leads to a much simpler, though surprisingly effective classifier in practice.

# Foundations of Deep Learning

## Covers

- ▶ Recap of linear models and discussion of stochastic optimization methods
- ▶ Basic understanding of modern neural networks and their applications in computer vision and natural language understanding,
- ▶ Understand the major technology trends driving Deep Learning,
- ▶ Be able to build, train and apply fully connected deep neural networks,
- ▶ Know how to implement efficient (vectorized) neural networks,
- ▶ Understand the key parameters in a neural network's architecture This course also teaches you how Deep Learning actually works, rather than presenting only a cursory or surface-level description.

# Foundations of Deep Learning

## Introductory example

### Visual product recommender

- ▶ Buy new articles (shoes, clothes,...)
- ▶ Look for an animals (Dog, Cat, ...)
- ▶ Write a program to automatically read someone's handwriting



Figure 18: Image from MNIST handwritten digit dataset

# Foundations of Deep Learning

Features are key to Machine Learning

## Problem

- ▶ Too many option online...
- ▶ Keywords are not very helpfull



Figure 19: A zero that's algorithmically difficult to distinguish from a six

- ▶ How can we represent features?
- ▶ Pb : learning **very** non-linear features

# Foundations of Deep Learning

Features are key to Machine Learning

- ▶ Object recognition
- ▶ Speech comprehension
- ▶ Automated translation
- ▶ ...

We don't know what program to write because we don't know how it's done by our brains. And even if we did know how to do it, the program might be horrendously complicated.

# Foundations of Deep Learning

The Mechanics of Machine Learning :Linear classifiers

## How to tackle these classes of problems?

We'll have to use a very different kind of approach :

**Learned by example**, not by formula

### Definition

Deep learning is a subset of a more general field of artificial intelligence called machine learning, which is predicated on this idea of learning from example

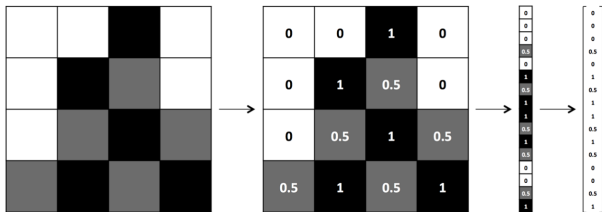


Figure 20: The process of vectorizing an image for a machine learning algorithm



# Foundations of Deep Learning

## Linear classifiers

### **What can a linear classifier represent?**

- ▶ OR
- ▶ AND
- ▶ How can we represent features?
- ▶ Pb : learning **very** non-linear features

# Foundations of Deep Learning

## Linear classifiers

### **What a linear classifier can't represent?**

- ▶ XOR
- ▶ How can we represent features?
- ▶ Pb : learning **very** non-linear features

# Foundations of Deep Learning

## Features extraction from an image

- ▶ How can we represent features?
- ▶ Pb : learning **very** non-linear features

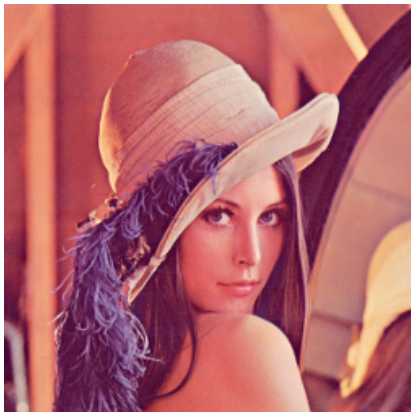


Figure 21: Features extraction

# Foundations of Deep Learning

## The Neuron



Figure 22: Features extraction